Уравнения математической физики

Найти решение краевой задачи для уравнения свободных колебаний струны:

1. $\frac{\partial^2 u}{\partial t^2} = 9\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = x(x-1)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $u(0,t) = u(1,t) = 0$

2. $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = x(x-3)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $u(0,t) = u(3,t) = 0$

3. $\frac{\partial^2 u}{\partial t^2} = 4\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = x(x-2)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $u(0,t) = u(2,t) = 0$

4. $\frac{\partial^2 u}{\partial t^2} = 25\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = x(x-4)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $u(0,t) = u(4,t) = 0$

5. $\frac{\partial^2 u}{\partial t^2} = 36\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = x(x-5)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $u(0,t) = u(5,t) = 0$

6. $\frac{\partial^2 u}{\partial t^2} = 49\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = x(x-6)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $u(0,t) = u(6,t) = 0$

7. $\frac{\partial^2 u}{\partial t^2} = 64\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = x(x-7)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $u(0,t) = u(7,t) = 0$

8. $\frac{\partial^2 u}{\partial t^2} = 81\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = x(x-8)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $u(0,t) = u(8,t) = 0$

9. $\frac{\partial^2 u}{\partial t^2} = 100\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = x(x-9)$, $\frac{\partial u}{\partial t}(x,0) = 0$, $u(0,t) = u(9,t) = 0$

10. $\frac{\partial^2 u}{\partial t^2} = 4\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = 0$, $\frac{\partial u}{\partial t}(x,0) = x(x-2)$, $u(0,t) = u(2,t) = 0$

11. $\frac{\partial^2 u}{\partial t^2} = 16\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = 0$, $\frac{\partial u}{\partial t}(x,0) = x(x-3)$, $u(0,t) = u(3,t) = 0$

12. $\frac{\partial^2 u}{\partial t^2} = 9\frac{\partial^2 u}{\partial x^2}$, если $u(x,0) = 0$, $\frac{\partial u}{\partial t}(x,0) = x(x-1)$, $u(0,t) = u(1,t) = 0$

13. $\dfrac{\partial^2 u}{\partial t^2} = 25\dfrac{\partial^2 u}{\partial x^2}$, если $u(x,0)=0$, $\dfrac{\partial u}{\partial t}(x,0) = x(x-4)$, $u(0,t)=u(4,t)=0$

14. $\dfrac{\partial^2 u}{\partial t^2} = 36\dfrac{\partial^2 u}{\partial x^2}$, если $u(x,0)=0$, $\dfrac{\partial u}{\partial t}(x,0) = x(x-5)$, $u(0,t)=u(5,t)=0$

15. $\dfrac{\partial^2 u}{\partial t^2} = 49\dfrac{\partial^2 u}{\partial x^2}$, если $u(x,0)=0$, $\dfrac{\partial u}{\partial t}(x,0) = x(x-6)$, $u(0,t)=u(6,t)=0$

16. $\dfrac{\partial^2 u}{\partial t^2} = 64\dfrac{\partial^2 u}{\partial x^2}$, если $u(x,0)=0$, $\dfrac{\partial u}{\partial t}(x,0) = x(x-7)$, $u(0,t)=u(7,t)=0$

17. $\dfrac{\partial^2 u}{\partial t^2} = 16\dfrac{\partial^2 u}{\partial x^2}$, если $u(x,0)=0$, $\dfrac{\partial u}{\partial t}(x,0) = x(x-8)$, $u(0,t)=u(8,t)=0$

18. $\dfrac{\partial^2 u}{\partial t^2} = 25\dfrac{\partial^2 u}{\partial x^2}$, если $u(x,0)=0$, $\dfrac{\partial u}{\partial t}(x,0) = x(x-9)$, $u(0,t)=u(9,t)=0$

19. $\dfrac{\partial^2 u}{\partial t^2} = 16\dfrac{\partial^2 u}{\partial x^2}$, если $u(x,0)=x(x-4)$, $\dfrac{\partial u}{\partial t}(x,0) = 0$, $u(0,t)=u(4,t)=0$

20. $\dfrac{\partial^2 u}{\partial t^2} = \dfrac{\partial^2 u}{\partial x^2}$, если $u(x,0)=x(x-3)$, $\dfrac{\partial u}{\partial t}(x,0) = 0$, $u(0,t)=u(3,t)=0$